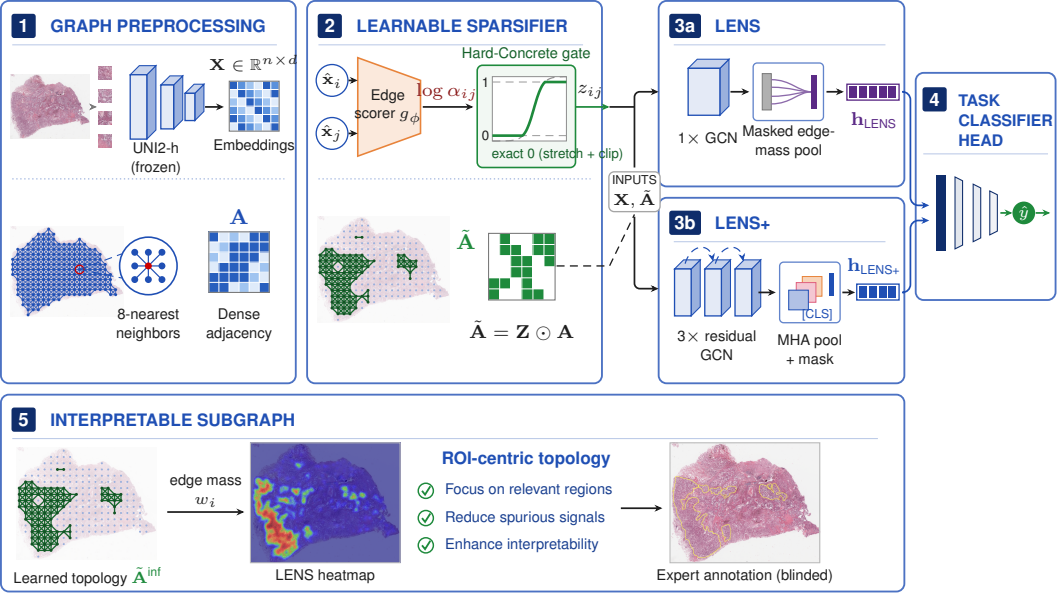

1 GRAPH PREPROCESSING
$\mathbf{X} \in \mathbb{R}^{n \times d}$
UNI2-h (frozen)
Embeddings
A
8-nearest neighbors
Dense adjacency
2 LEARNABLE SPARSIFIER
Edge scorer $g_\phi$
$\log \alpha_{ij}$
Hard-Concrete gate
1
0
exact 0 (stretch + clip)
$z_{ij}$
$\tilde{\mathbf{A}}$
$\tilde{\mathbf{A}} = \mathbf{Z} \odot \mathbf{A}$
INPUTS
$\mathbf{X}, \tilde{\mathbf{A}}$
3a LENS
1× GCN
Masked edge-mass pool
$\mathbf{h}_{\text{LENS}}$
3b LENS+
3× residual GCN
[CLS]
MHA pool + mask
$\mathbf{h}_{\text{LENS+}}$
4 TASK CLASSIFIER HEAD
$\hat{y}$
5 INTERPRETABLE SUBGRAPH
Learned topology $\tilde{\mathbf{A}}^{\text{inf}}$
edge mass $w_i$
LENS heatmap
ROI-centric topology
Focus on relevant regions
Reduce spurious signals
Enhance interpretability
Expert annotation (blinded)